A tangent-groupoid formulation of the quantum construction of Newton motion

Result, 2026-10-27. The tangent groupoid carries a continuous field of observable algebras whose positive-parameter fibers are quantum and whose zero fiber is classical. After fixing an action unit S_* and setting $h = \epsilon S_*$, the finite quantum products and classical initial-value construction of the first document define compatible states on these fibers. On every compact time interval, their values on smooth Weyl observables converge to the classical point state with an explicit $O(h + |\pi|)$ bound, for **every joint schedule** with $h, |\pi| \to 0$.

The result is a state-level limit for the specified coherent preparations and smooth potentials with bounded derivatives of order at least two, including the third derivative used for the rate. The quartic extension in the first document uses cutoff Hamiltonians; it does not establish this estimate for uncut quartic quantum evolution. The result neither makes the finite-product propagator uniformly close to the exact quantum propagator as $h \to 0$, nor makes a quantum superposition a single classical trajectory. The groupoid parameter is a geometric deformation parameter until the identification $h = \epsilon S_*$ is chosen. The time mesh is an additional scale.

The didactic guide introduces the Dimock–Balaban comparison and the point-particle examples. Its Dirichlet hard-wall calculation is a separate construction; the whole-line groupoid and state theorem below do not automatically cover that boundary problem.

The groupoid and strict quantization are established: in particular, Cariñena, Clemente-Gallardo, Follana, Gracia-Bondía, Rivero and Várilly, Connes' Tangent Groupoid and Strict Quantization (abstract and the prior repository companion), already includes the user's work. Landsman–Ramazan's continuous-field construction (abstract) supplies the general Lie-groupoid setting. The following state estimate applies the first document's finite-slice bounds to that established structure; no literature priority claim is made.

1. What the parameter joins

For $M = \mathbb{R}$, the tangent groupoid is

$$\mathbb{T}M = (M \times M \times (0,1]) \;\sqcup\; (TM \times \{0\}). \tag{1}$$

At $\epsilon > 0$, pair arrows compose as $(x, y, \epsilon)(y, z, \epsilon) = (x, z, \epsilon)$; at zero, $(q, v, 0)(q, w, 0) = (q, v + w, 0)$. A local chart near the zero fiber is

$$(q, v, \epsilon) \longmapsto \big(q + \epsilon v/2,\; q - \epsilon v/2,\; \epsilon\big). \tag{2}$$

Thus $v = (x - y)/\epsilon$ is a **tangent coordinate**, not automatically a physical velocity. A physical time interpretation requires a separate identification of units and scaling.

The groupoid C^*-algebra is a continuous field with fibers

$$\mathcal{A}_\epsilon \simeq \mathcal{K}(L^2(\mathbb{R})) \quad (\epsilon > 0), \qquad \mathcal{A}_0 \simeq C_0(T^*\mathbb{R}). \tag{3}$$

At zero, Fourier transformation takes tangent-vector convolution to pointwise multiplication of functions of (q, p). The positive fibers are abstractly isomorphic, but the continuous field specifies how families of their elements approach a classical observable.

Choose an action unit $S_* > 0$ and set

$$h = \epsilon S_*. \tag{4}$$

For $a \in C_c^\infty(T^*\mathbb{R})$, Weyl quantization has kernel

$$K_\epsilon^a(x, y) = \frac{1}{2\pi\epsilon S_*} \int_{\mathbb{R}} e^{i(x-y)p/(\epsilon S_*)} a\left(\frac{x+y}{2}, p\right) dp. \tag{5}$$

In chart (2),

$$\epsilon K_\epsilon^a(q+\epsilon v/2, q-\epsilon v/2) = \frac{1}{2\pi S_*}\int e^{ivp/S_*} a(q,p)dp. \tag{6}$$

The rescaled kernel is the inverse Fourier transform of the zero-fiber symbol. The established strict-quantization construction makes $\epsilon \mapsto \mathrm{Op}^W_{\epsilon S_*}(a)$, together with a at zero, a continuous section. Equation (6) displays the chart and normalization in this flat case; it is not a new proof of the general continuous-field theorem.

2. The dynamics and the unbounded objects

For the potential and hypotheses of the first document define

$$H_h = P^2/(2m) + V(Q), \qquad U_h(t) = e^{-itH_h/h}. \tag{7}$$

The H_h are unbounded operators and the $U_h(t)$ are generally multipliers, not elements of the compact-operator fibers. At $h > 0$ their adjoint action defines automorphisms of the compact operators:

$$\alpha_h^t(A) = U_h(t)^* A U_h(t). \tag{8}$$

The first document constructs the unique classical phase-space flow $\Phi_t(q,p)$ before it is used here. On the zero fiber put

$$\alpha_0^t(a) = a \circ \Phi_t. \tag{9}$$

This is classical Hamiltonian evolution with $H_{\mathrm{cl}} = p^2/(2m) + V(q)$.

For $a \in C_c^\infty(T^*\mathbb{R})$, Weyl–Egorov compatibility for the smooth subquadratic potentials of the first document gives, for fixed T,

$$\sup_{|t|\le T} \|\alpha_h^t(\mathrm{Op}_h^W(a)) - \mathrm{Op}_h^W(a\circ\Phi_t)\| \le C_{a,T} h^2. \tag{10}$$

Here is the relevant estimate. The finite-time flow has bounded derivatives on compact sets under the stated hypotheses, so $a_s = a \circ \Phi_s$ is smooth, compactly supported, and has uniform symbol bounds for $|s| \le T$. The kinetic commutator is exact. The potential-commutator remainder after removing its classical first-order term has kernel

$$\frac{i}{h}\left[V(x) - V(y) - (x-y)V'\!\left(\frac{x+y}{2}\right)\right] K_h^{a_s}(x,y). \tag{10a}$$

The bracket is bounded by $\|V'''\|_\infty |x-y|^3/24$. Rapid decay of the Fourier transform of a_s in its momentum variable, the change $x - y = hz$, and Schur's test bound the operator norm by $C_{a,T}h^2$. Integrating this commutator remainder by Duhamel's formula gives (10). This uses the standard Weyl calculus and the flow already constructed in the first document; it is not being used to prove that flow exists.

3. A quantitative state on every fiber

Use the coherent packet $\psi_h^{q_0,p_0}$ of equation (16) in the first document. For a partition π, let $\psi_{h,\pi,j}$ be the state after its first j kick–drift cells, and let $z_{h,\pi}(t_j) = (q_j, p_j)$ be its means. Denote the constructed classical state by $z(t) = \Phi_t(q_0,p_0)$. Define

$$I_a = (2\pi)^{-2}\int_{\mathbb{R}^2} |\widehat{a}(\xi,\eta)|\,(|\xi| + m\omega|\eta|)\,d\xi\,d\eta < \infty. \tag{11}$$

Here $\widehat{a}$ is the ordinary Fourier transform in (q,p). Also put

$$J_a = \frac{1}{2(2\pi)^2}\int_{\mathbb{R}^2} |\widehat{a}(\xi,\eta)|\,(|\xi| + m\omega|\eta|)^2\,d\xi\,d\eta < \infty. \tag{11a}$$

Using the exact Fourier representation of Weyl quantization,

$$\mathrm{Op}_h^W(a) = (2\pi)^{-2} \int \widehat{a}(\xi,\eta) e^{i(\xi Q+\eta P)} d\xi\, d\eta, \tag{12}$$

set $A = \xi(Q-q_j)+\eta(P-p_j)$. The scalar phase $e^{i(\xi q_j+\eta p_j)}$ factors out, $\langle A\rangle_j = 0$, and spectral calculus gives $|\langle e^{iA}\rangle_j - 1| \le \frac{1}{2}\langle A^2\rangle_j$. Consequently,

$$|\langle \mathrm{Op}_h^W(a)\rangle_j - a(q_j,p_j)| \le J_a \left[\mathrm{sd}(Q)_j + \frac{\mathrm{sd}(P)_j}{m\omega}\right]^2. \tag{13}$$

Indeed $\langle A^2\rangle_j = \|A\psi_{h,\pi,j}\|^2$ and the triangle inequality bounds $\|A\psi_{h,\pi,j}\|$ by the weighted sum of the two standard deviations. No Gaussian-shape assumption is needed after the first step. Fourier inversion also gives $|a(z)-a(z')| \le I_a \|z-z'\|_\omega$.

The first document's estimates (19) and (26) therefore give, at every partition vertex $t_j \in [0,T]$,

$$\left|\langle \psi_{h,\pi,j}, \mathrm{Op}_h^W(a)\psi_{h,\pi,j}\rangle - a(z(t_j))\right| \le J_a C^2 e^{2\beta T} h + I_a G_T (C_\delta |\pi| + C_h h). \tag{14}$$

The constants $C, \beta, G_T, C_\delta, C_h$ are explicitly defined in that document and depend on the common h_0, T, V, m, ω and initial data, not on h or the number of cells. No differentiability of an unbounded quantum expectation is used in this step.

With $h = \epsilon S_*$, choose any $\alpha > 0$ and a time unit $T_* > 0$ so that $|\pi_\epsilon| \le T_* \epsilon^\alpha$. Equation (14) yields

$$\langle \mathrm{Op}^W_{\epsilon S_*}(a)\rangle_{\pi_\epsilon} \longrightarrow a(\Phi_t(q_0,p_0)) \quad \text{at vertices tending to } t, \tag{15}$$

with error $O(\epsilon+\epsilon^\alpha)$ at vertices equal to t. Such a vertex can be inserted in each partition. There is no constraint here on $|\pi_\epsilon|/h$.

This means that the vector states on the positive fibers approach the point-evaluation state on the zero fiber, for the specified family of preparations. For an arbitrary continuous section A, approximate $A(0) \in C_0(T^*\mathbb{R})$ uniformly by compactly supported smooth symbols. Continuity of the field and the norm-one bound on states extend (15) from these symbols to A. At fixed h, taking the mesh limit recovers the exact unitary state as in the first document. Both routes thus reach the same zero-fiber state; equality or closeness of the two quantum vectors along an arbitrary joint schedule has **not** been established.

4. A precise two-scale test on Euclidean increments

The preceding theorem concerns real-time operator states. It does not claim that Euclidean sample paths are real-time trajectories. The following free Gaussian calculation shows why the groupoid coordinate, the action parameter and the duration must not be conflated.

Consider a drifted free reference increment

$$Y - X = u\delta + \sqrt{\frac{h\delta}{m}}\, Z, \qquad Z \sim N(0,1). \tag{16}$$

Set $h = \epsilon S_*$ and $\delta = T_* \epsilon^\alpha$ with $\alpha > 0$. The physical-time difference quotient is

$$\frac{Y-X}{\delta} = u + \sqrt{\frac{S_*}{mT_*}}\, \epsilon^{(1-\alpha)/2} Z, \tag{17}$$

whereas the tangent-groupoid chart variable is

$$v_\epsilon = \frac{Y-X}{\epsilon} = uT_*\epsilon^{\alpha-1} + \sqrt{\frac{S_*T_*}{m}}\, \epsilon^{(\alpha-1)/2} Z. \tag{18}$$

- If $0 < \alpha < 1$, the physical-time quotient concentrates at u, but v_ϵ is not tight.

- If $\alpha = 1$, v_ϵ converges in law to a nondegenerate Gaussian of mean uT_* and variance S_*T_*/m.
- If $\alpha > 1$, $v_\epsilon \to 0$ in probability, while the physical-time quotient is not tight.

This is an exact diagnostic for the Euclidean Gaussian reference, not a probability law for real-time Feynman paths. It also does not contradict (15): a groupoid observable is obtained after the Fourier and operator construction, whereas (18) examines a single sampled Euclidean increment.

5. Where blocking sits in the groupoid picture

For each $\epsilon > 0$ the first document supplies a complete tower $\rho_{\epsilon,n}$ of normalized conditional densities on dyadic time samples. The relation

$$\rho_{\epsilon,n} = \mathbb{E}_{0,\epsilon}[\rho_{\epsilon,n+1} \mid \mathcal{F}_n] \tag{19}$$

is exact and retains all generated interactions. The integer n counts an integration/coarse-graining operation; ϵ labels the deformation field (4). The conditional maps are generally not invertible, so they are not automatically arrows of the tangent groupoid.

Equation (15) connects these scales for the real-time **observable states** even when they are tied by a single chosen schedule. It does not follow from the $1/Z_\epsilon$ Euclidean bound, which need not be uniform in ϵ, and it does not promote Euclidean minimization into real-time Newton dynamics.

The zoom action on the tangent groupoid is a geometric rescaling of near-diagonal arrows. It is useful for organizing a comparison with RG, but it is not by itself an integration of fluctuations or a theorem that the physical action constant runs. Likewise, the real-time Hamilton principal action composes by stationary elimination on a specified noncaustic branch; infimal convolution is appropriate for a genuine minimizing Euclidean problem or a separately proved special case. The older groupoid note is to be read with these qualifications.

6. Consequence for STATE

The constructed $0 + 1$ quantum family, the semiclassical Newton flow and the user's tangent-groupoid framework now meet in a concrete state theorem with a joint mesh/action error bound. The strong operator product limit remains a fixed-h theorem; the Euclidean density tower and the real-time coherent states have distinct roles. No positive universal action scale or four-dimensional Yang–Mills result follows.

Targeted advisor check (GPT-6.1 Sol, 2026-10-27). An internal mathematical review checked the chart and Fourier normalizations, the state and Gaussian-scale bounds, the extension to continuous sections, and the estimate (10). It is not independent peer review of the underlying continuous-field theorem or of the literature. The centered $O(h)$ state improvement in §3 is a subsequent written derivation and was not part of that earlier check.